

3-SCRAMBLED CANTOR SETS

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ABSTRACT. We provide a homeomorphism of Cantor space that has a single uncountable set which is d -scrambled for all $d \geq 3$, but which does not have a 3-scrambled Cantor set.

INTRODUCTION

Given a metric space X , $f: X \rightarrow X$, and $d \geq 2$, a sequence $x \in X^d$ is *proximal* if $\liminf_{n \rightarrow \infty} \max_{i < j < d} d_X(f^n(x(i)), f^n(x(j))) = 0$, *separated* if $\limsup_{n \rightarrow \infty} \min_{i < j < d} d_X(f^n(x(i)), f^n(x(j))) > 0$, and *Li–Yorke* if it is proximal and separated. A set $Y \subseteq X$ is *d-scrambled* if every injective sequence in Y^d is Li–Yorke. We say that Y is $(<\mathbb{N})$ -*scrambled* if it is d -scrambled for all $d \geq 2$.

In [GGM21], it was shown that uncountable 2-scrambled sets yield 2-scrambled Cantor sets in Polish dynamical systems. Here we show:

Theorem 1 (ZFC). *There is a homeomorphism of $2^{\mathbb{N}}$ with an uncountable $(<\mathbb{N})$ -scrambled set but no 3-scrambled Cantor set.*

A *graph* on X is an irreflexive symmetric set $G \subseteq X \times X$. We say that Y is a *G-clique* if its distinct points are G -related, and *G-independent* if its points are not G -related. Let $\text{CLQ}_G^{\geq 3}$ and $\text{IND}_G^{\geq 3}$ denote the sets of injective finite sequences of length at least three whose images have these properties. Let LY_f denote the set of finite injective Li–Yorke sequences. A *homomorphism* from a relation R on X to a relation S on Y is a function $\pi: X \rightarrow Y$ whose compositions with sequences in R are in S . Such a function is a *homomorphism* from (R, R') to (S, S') if it is also a homomorphism from R' to S' . The *saturation* of a set $Y \subseteq X$ under a bijection $T: X \rightarrow X$ is $[Y]_T = \bigcup_{n \in \mathbb{Z}} T^n(Y)$. A subset of a topological space is F_σ if it is a countable union of closed sets. The primary new technical result underlying our proof of Theorem 1 is:

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Theorem 2 (ZF + DC). *Suppose that G is an F_σ graph on $2^\mathbb{N}$. Then there is a homeomorphism $T: 2^\mathbb{N} \rightarrow 2^\mathbb{N}$ for which there is a continuous injective homomorphism $\pi: 2^\mathbb{N} \rightarrow 2^\mathbb{N}$ from $(\text{CLQ}_G^{\geq 3}, \text{IND}_G^{\geq 3})$ to $(\text{LY}_T, \sim \text{LY}_T)$ such that $|\sim[\pi(2^\mathbb{N})]_T| = 1$.*

In §1, we establish Theorem 2. In §2, we obtain Theorem 1 and a related independence phenomenon as corollaries.

1. EMBEDDING HYPERGRAPHS

Define $S: 2^\mathbb{Z} \rightarrow 2^\mathbb{Z}$ by $S(c)(i) = c(i+1)$ for all $i \in \mathbb{Z}$ and $c \in 2^\mathbb{Z}$. Let SEP_f denote the set of finite injective separated sequences. Theorem 2 follows from the usual topological characterization of $2^\mathbb{N}$ and:

Theorem 1.1 (ZF + DC). *Suppose that G is an F_σ graph on $2^\mathbb{N}$. Then there are an S -invariant perfect set $F \subseteq 2^\mathbb{Z}$, whose finite sequences are proximal, and a continuous injective homomorphism $\pi: 2^\mathbb{N} \rightarrow F$ from $(\text{CLQ}_G^{\geq 3}, \text{IND}_G^{\geq 3})$ to $(\text{SEP}_S, \sim \text{SEP}_S)$ such that $F \setminus [\pi(2^\mathbb{N})]_S = \{0^\mathbb{Z}\}$.*

Proof. For all $s \in 2^{<\mathbb{N}}$, define $\mathcal{N}_s = \{c \in 2^\mathbb{N} \mid \forall i < |s| \ s(i) = c(i)\}$. For all $S \subseteq 2^{<\mathbb{N}}$, define $\mathcal{N}_S = \bigcup_{s \in S} \mathcal{N}_s$. For all $d \geq 3$, $\ell \in \mathbb{N}$, and $p: 2^\ell \rightarrow d$, define $\ell_p = \ell$ and $\mathcal{N}_p = \prod_{i < d} \mathcal{N}_{p^{-1}(\{i\})}$. We say that $p: 2^\ell \rightarrow d$ is a *near bijection* if it is surjective and there is a unique $k_p < d$ for which $|p^{-1}(\{k_p\})| > 1$. For all $k \in d \setminus \{k_p\}$, set $\{p^{-1}(k)\} = p^{-1}(\{k\})$.

Fix an increasing sequence of closed graphs G_n on $2^\mathbb{N}$ for which $G = \bigcup_{n \in \mathbb{N}} G_n$. Define $G_{m,n} = \{(c, c') \in 2^\mathbb{N} \times 2^\mathbb{N} \mid G_n \cap (\mathcal{N}_{c \upharpoonright m} \times \mathcal{N}_{c' \upharpoonright m}) \neq \emptyset\}$, and let $P_{d,m,n}$ denote the set of all near bijections $p: 2^m \rightarrow d$ such that $k_p = 0 = p(0^m)$ and $\mathcal{N}_{p^{-1}(j)} \times \mathcal{N}_{p^{-1}(k)} \subseteq G_{m,n}$ for all $0 < j < k < d$. Fix $I \subseteq \mathbb{N} \setminus \{0\}$ and a bijection $\phi: I \rightarrow \mathbb{N} \cup \bigcup_{d \geq 3, m, n \in \mathbb{N}} \{d\} \times \{m\} \times \{n\} \times P_{d,m,n}$ such that $I \cap [i, 2i + w(i)] = \{i\}$ for all $i \in I$, where $w(i) = 0$ for $i \in \phi^{-1}(\mathbb{N})$ and $w(i) = (\phi(i)(0) - 2)(\phi(i)(2) + 1)$ for $i \in I \setminus \phi^{-1}(\mathbb{N})$. Define a continuous function $\pi: 2^\mathbb{N} \rightarrow 2^\mathbb{Z}$ by $\pi(c)(i) = 1$ if and only if

- (1) $i = 0$;
- (2) $i \in I$, $\phi(i) \in \mathbb{N}$, and $c(\phi(i)) = 1$; or
- (3) $\exists d \geq 3 \exists 0 < k < d \exists m, n \in \mathbb{N} \exists p \in P_{d,m,n}$
 $((d, m, n, p) = \phi(i - (k - 1)(n + 1)) \text{ and } p(c \upharpoonright \ell_p) = k)$.

Lemma 1.2. π is injective.

Proof. Suppose that $c, c' \in 2^\mathbb{N}$ are distinct. Then there exists $n \in \mathbb{N}$ with $c(n) \neq c'(n)$. Fix $i \in I$ for which $\phi(i) = n$. As $\pi(c)(i) = c(n)$ and $\pi(c')(i) = c'(n)$, it follows that $\pi(c)(i) \neq \pi(c')(i)$, so $\pi(c) \neq \pi(c')$. $\square$

The *product metric* on $2^\mathbb{Z}$ is given by $\rho(c, c') = \sum_{i \in \mathbb{Z}} |c(i) - c'(i)|/2^{|i|}$ for all $c, c' \in 2^\mathbb{Z}$. Define $N_i = [i, i + w(i)]$ for all $i \in I$.

Lemma 1.3. π is a homomorphism from $\text{CLQ}_G^{\geq 3}$ to SEP_S .

Proof. Suppose that $c \in (2^{\mathbb{N}})^d$ is in $\text{CLQ}_G^{\geq 3}$. Fix $n \in \mathbb{N}$ sufficiently large that $c(j) G_n c(k)$ for all $j < k < d$. By permuting the coordinates of c , we can assume that if $c(k) = 0^{\mathbb{N}}$ for some $k < d$, then $k = 0$. Suppose that $m \in \mathbb{N}$ is sufficiently large that $\text{proj}_{2^m} \circ c$ is injective, $c(k) \upharpoonright m \neq 0^m$ for all $0 < k < d$, and $d < 2^m$. Let $p \in P_{d,m,n}$ be the near bijection extending $(\text{proj}_{2^m} \circ c)^{-1}$ for which $k_p = 0$ and $\ell_p = m$. Fix $i \in I$ such that $\phi(i) = (d, m, n, p)$ and observe that $\text{proj}_{2^{N_i}} \circ \pi \circ c$ is injective, so $\min_{j < k < d} \rho(S^i(\pi(c(j))), S^i(\pi(c(k)))) \geq 1/2^{(d-2)(n+1)}$. $\square$

Lemma 1.4. π is a homomorphism from $\text{IND}_G^{\geq 3}$ to $\sim\text{SEP}_S$.

Proof. Suppose that $c \in (2^{\mathbb{N}})^d$ and $\pi \circ c$ is separated. Fix $\epsilon > 0$ such that $I' = \{i' \in \mathbb{N} \mid \min_{j < k < d} \rho(S^{i'}(\pi(c(j))), S^{i'}(\pi(c(k)))) \geq \epsilon\}$ is infinite, as well as $r \in \mathbb{N}$ such that $\sum_{|i| > r} 1/2^{|i|} < \epsilon$. By throwing out finitely many elements of I' , we can assume that for all $i' \in I'$, there is at most one $i \in I$ such that $[i' - r, i' + r] \cap N_i \neq \emptyset$. Our choice of r then yields a unique such $i \in I$, in which case $\text{proj}_{2^{N_i}} \circ \pi \circ c$ is injective. Set $(e, m, n, p) = \phi(i)$. As $e \geq 3$, there exist $j < k < d$ for which $(d - 2)(n + 1) \leq 2r$, and $c(j) G_{m,n} c(k)$. By passing to an infinite subset of I' , we can assume that the values of j , k , and n are independent of $i' \in I'$. Then there are unboundedly many values of m corresponding to the remaining $i' \in I'$, since otherwise there are only finitely many possibilities for both e and p , and therefore only finitely many possibilities for i , despite the fact that the distance between any two elements of I' corresponding to the same value of i is at most $2r$. It follows that $c(j) G_n c(k)$, so $c(j) G c(k)$, thus $c \notin \text{IND}_G^{\geq 3}$. $\square$

The closure F of the S -saturation of $\pi(2^{\mathbb{N}})$ is clearly perfect.

Lemma 1.5. The sequence $0^{\mathbb{Z}}$ is the unique element of F not in $[\pi(2^{\mathbb{N}})]_S$.

Proof. If $(c_n)_{n \in \mathbb{N}}$ is a sequence of elements of $2^{\mathbb{N}}$ and $(k_n)_{n \in \mathbb{N}}$ is a strictly increasing sequence of elements of $\mathbb{N}$, then $S^{-k_n}(\pi(c_n)) \rightarrow 0^{\mathbb{Z}}$. And if $S^{k_n}(\pi(c_n))$ converges to a sequence c other than $0^{\mathbb{Z}}$, then $c(n) = 1$ for a unique $n \in \mathbb{Z}$, so it is in the S -orbit of $\pi(0^{\mathbb{N}})$. $\square$

Lemma 1.5 implies that the union of the supports of finitely many elements of F is contained in a finite union of translates of $\{0\} \cup \bigcup_{i \in I} N_i$. But such sets necessarily have arbitrarily long gaps, so every finite sequence of elements of F is proximal. $\square$

Proposition 1.6 (ZF + DC). Suppose that $d \geq 3$, X and Y are Polish spaces, G is a Borel graph on X , $T: Y \rightarrow Y$ is a Borel automorphism,

$B \subseteq Y$ is an uncountable Borel d -scrambled set, $\pi: X \rightarrow Y$ is an injective Borel homomorphism from $\text{IND}_G^{\geq 3}$ to $\sim\text{LY}_T$, and $[\pi(X)]_T$ is co-countable. Then G has a perfect clique.

Proof. Fix $i \in \mathbb{Z}$ for which $B \cap (T^i \circ \pi)(X)$ is uncountable. The perfect set theorem and Galvin's Theorem (see, for example, [Kec95, Theorems 13.6 and 19.7]) then yield a perfect set $P \subseteq (T^i \circ \pi)^{-1}(B)$ that is either a G -clique or G -independent. But G -independence would contradict the fact that B is d -scrambled. $\square$

2. d -SCRAMBLED SETS

We begin this section with the proof of our primary result:

Proof (of Theorem 1). By [KV12, Theorem 2.1] and a simple Baire category argument, there is an F_σ graph G on $2^\mathbb{N}$ with an uncountable clique $Y \subseteq 2^\mathbb{N}$ but no perfect clique. By Theorem 2, there is a continuous injective homomorphism $\pi: 2^\mathbb{N} \rightarrow 2^\mathbb{N}$ from $(\text{CLQ}_G^{\geq 3}, \text{IND}_G^{\geq 3})$ to $(\text{LY}_T, \sim\text{LY}_T)$ with $|\sim[\pi(2^\mathbb{N})]_T| = 1$. Then $\pi(Y)$ is $(<\mathbb{N})$ -scrambled, but Proposition 1.6 rules out a perfect 3-scrambled set. $\square$

Let $\mathfrak{c}$ denote the cardinality of the continuum.

Theorem 2.1 (ZFC + $\neg\text{CH}$). *It is consistent that there is a homeomorphism $T: 2^\mathbb{N} \rightarrow 2^\mathbb{N}$ with a $(<\mathbb{N})$ -scrambled set of cardinality $\mathfrak{c}$ but no perfect 3-scrambled set.*

Proof. By [She99, Theorem 1.13], we can assume that there is a F_σ graph G on $2^\mathbb{N}$ with a clique of cardinality $\mathfrak{c}$ but no perfect clique. The proof of Theorem 1 therefore yields the desired result. $\square$

A *hypergraph* is a permutation-invariant set of injective sequences.

Theorem 2.2 (ZFC + $\neg\text{CH}$). *Suppose that $d \geq 3$. Then it is consistent that every Borel function on a Polish space with a d -scrambled set of cardinality $\aleph_2$ has a d -scrambled perfect set.*

Proof. By [KS03, Proposition 3.4], we can assume that every Borel d -ary hypergraph on a Polish space with a clique of cardinality $\aleph_2$ has a perfect clique. But d -ary Li–Yorke-ness is a Borel hypergraph. $\square$

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